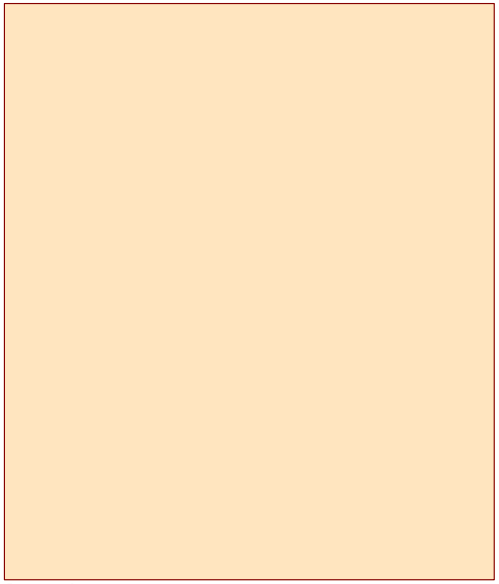
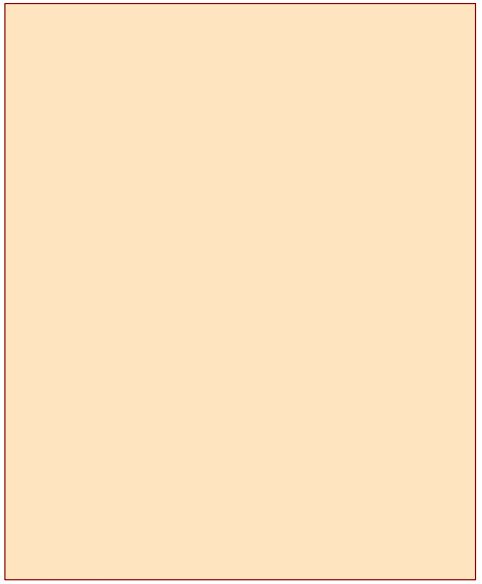
INDEX

escSheet



File: escSheet.kicad_sch

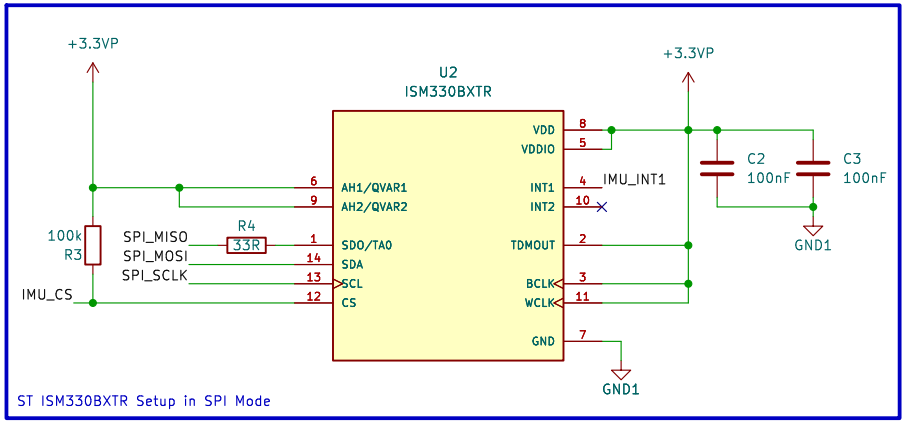
SensorSheet



File: SensorSheet.kicad_sch

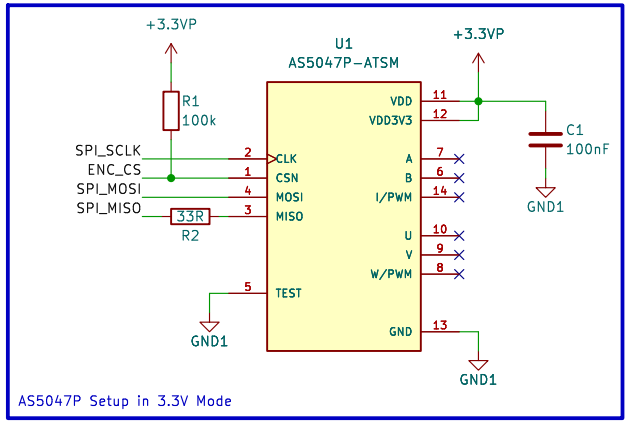
This Design is divided into two PCBs, connected with a 8-Pin connector Cable

IMU for Vibration Measurement Caused by Torque Ripple



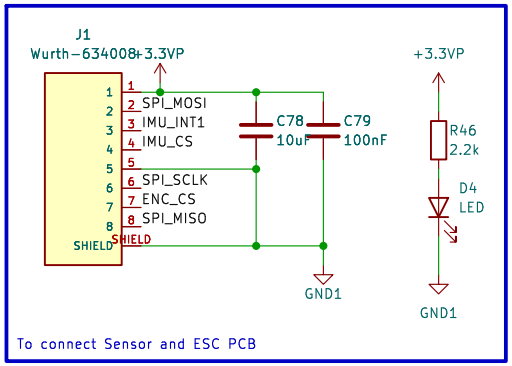
ST ISM330BXTR Setup in SPI Mode

Magnetic Encoder for Position Measurement
Place in Rotor Center with Diametral Magnet



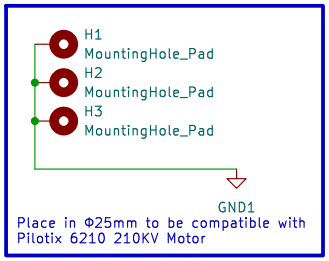
AS5047P Setup in 3.3V Mode

8-Pin 2.54mm Connector



To connect Sensor and ESC PCB

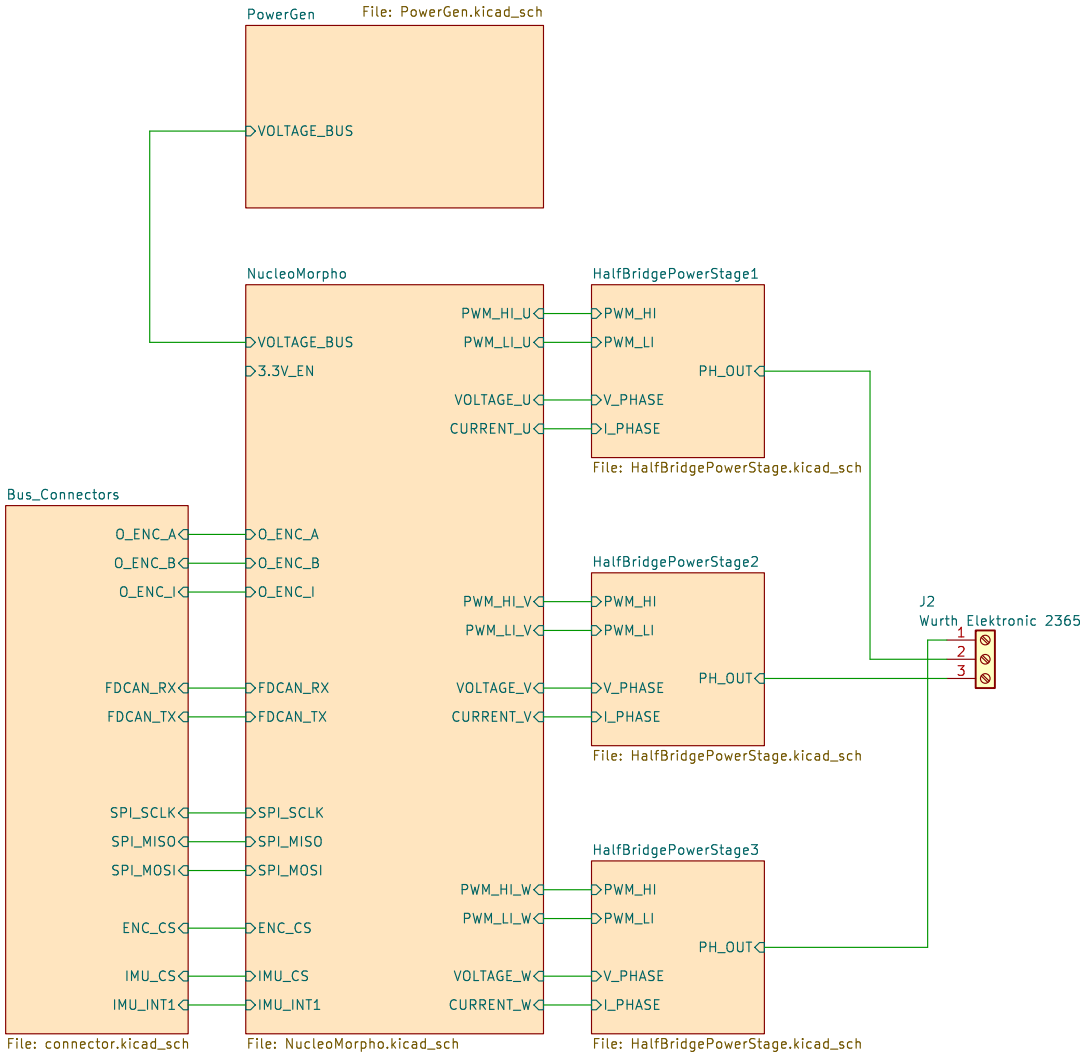
M4 Mounting Holes



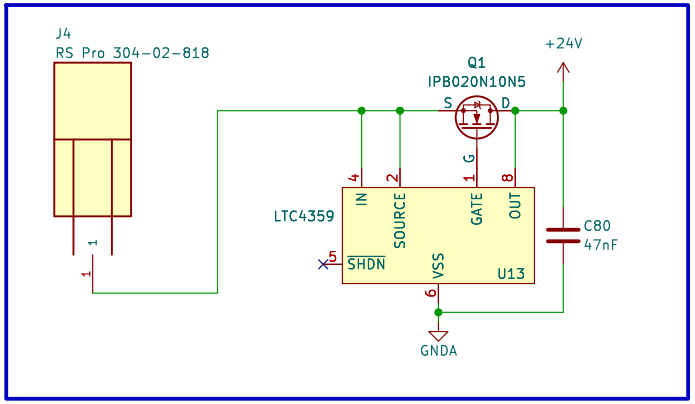
Place in Φ25mm to be compatible with Pilotix 6210 210KV Motor

330hm Resistors slow down dU/dt of MISO
Place close to driver

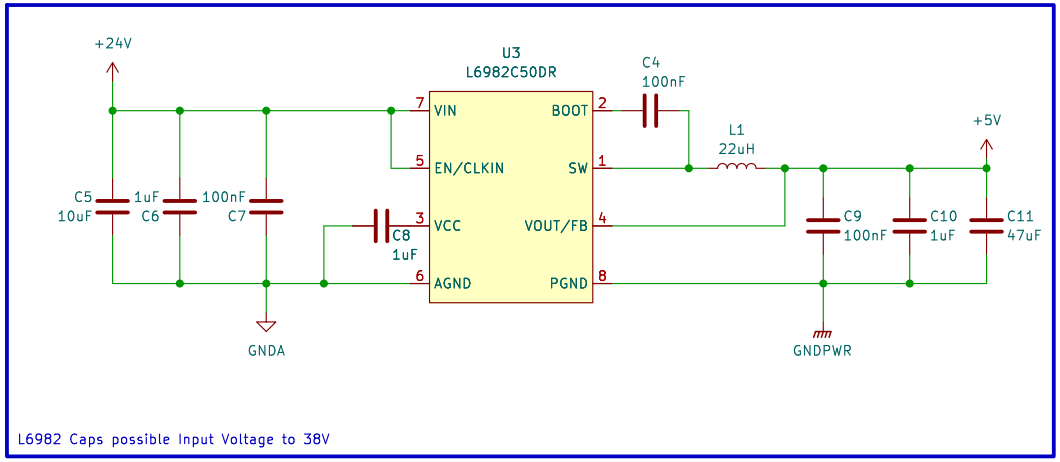
Pull-Ups on CS Pins


PowerGen
File: PowerGen.kicad_sch
VOLTAGE_BUS
NucleoMorpho
VOLTAGE_BUS
3.3V_EN
PWM_HI_U
PWM_LI_U
VOLTAGE_U
CURRENT_U
HalfBridgePowerStage1
PWM_HI
PWM_LI
PH_OUT
V_PHASE
I_PHASE
File: HalfBridgePowerStage.kicad_sch
Bus_Connectors
O_ENC_A
O_ENC_B
O_ENC_I
FDCAN_RX
FDCAN_TX
SPI_SCLK
SPI_MISO
SPI_MOSI
ENC_CS
IMU_CS
IMU_INT1
File: connector.kicad_sch
File: NucleoMorpho.kicad_sch
HalfBridgePowerStage2
PWM_HI_V
PWM_LI_V
VOLTAGE_V
CURRENT_V
File: HalfBridgePowerStage.kicad_sch
HalfBridgePowerStage3
PWM_HI_W
PWM_LI_W
VOLTAGE_W
CURRENT_W
File: HalfBridgePowerStage.kicad_sch
J2
Wurth Elektronic 2365
1
2
3

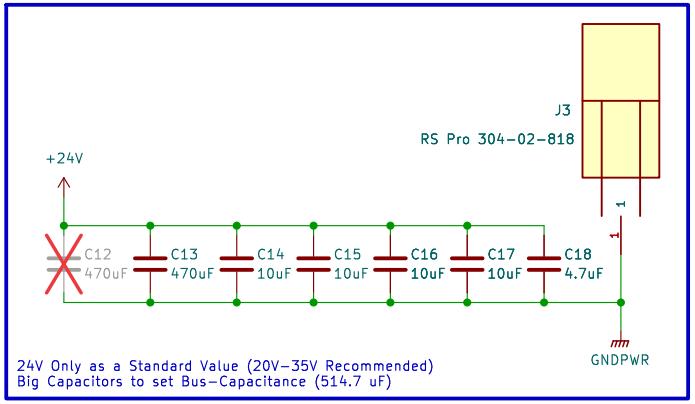
Ideal Diode Reverse Polarity Protection



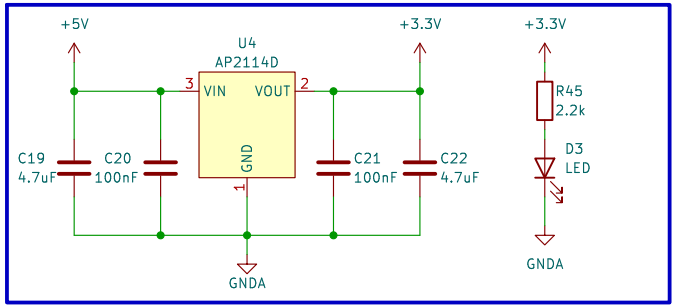
Generation +5V -- Uses ST Buck-Converter



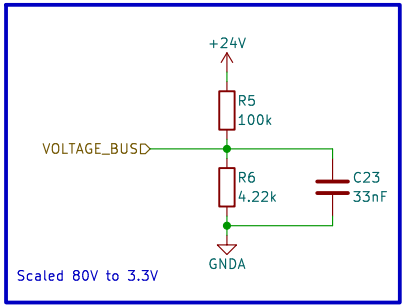
DC-Link Capacitor Bank



Low Noise Generation +3.3V -- Uses LDO



DC-Bus Voltage Sensing

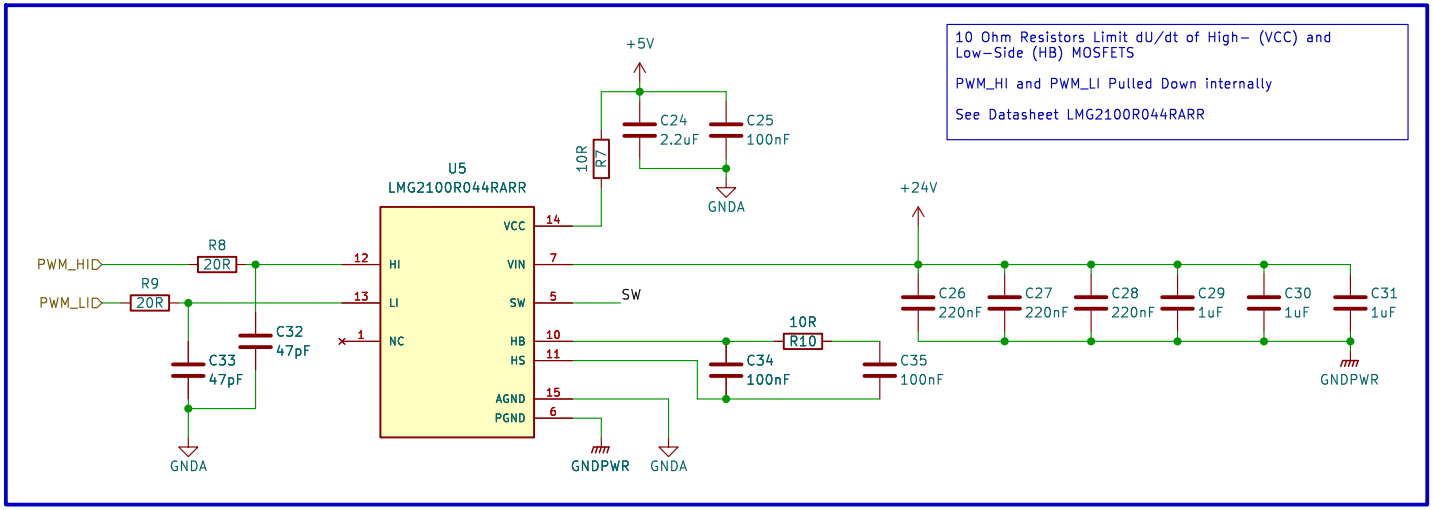


Bus Capacitance Calculation:
$C_{Bulk,min} > I_{ripple} / f_{PWM} / V_{ripple}$
$=> C_{Bulk,min} > p_{ripple,I} * I_{Motor} / (f_{PWM} * p_{ripple,V} * V_{Supply})$

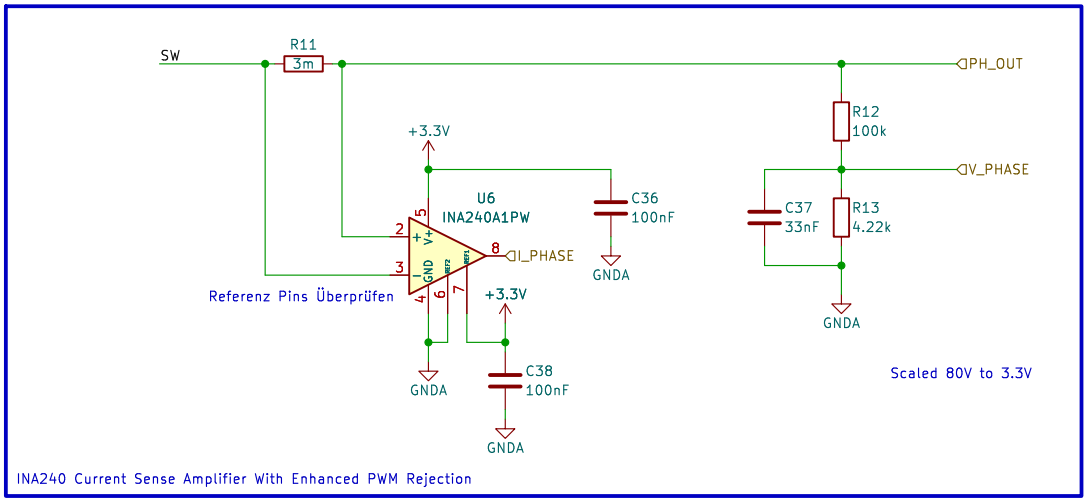
For 35A, 35V Input and 25kHz Switching Frequency
(using suggested Variances from TI Application Note):
$C_{Bulk,min} > 10\% * 15A / (25kHz * 1\% * 35V) = 172uF$

Scale Factor k = 3
$C_{Bulk} = k * C_{Bulk,min}$
$C_{Bulk} = 3 * 172uF = 514uF$

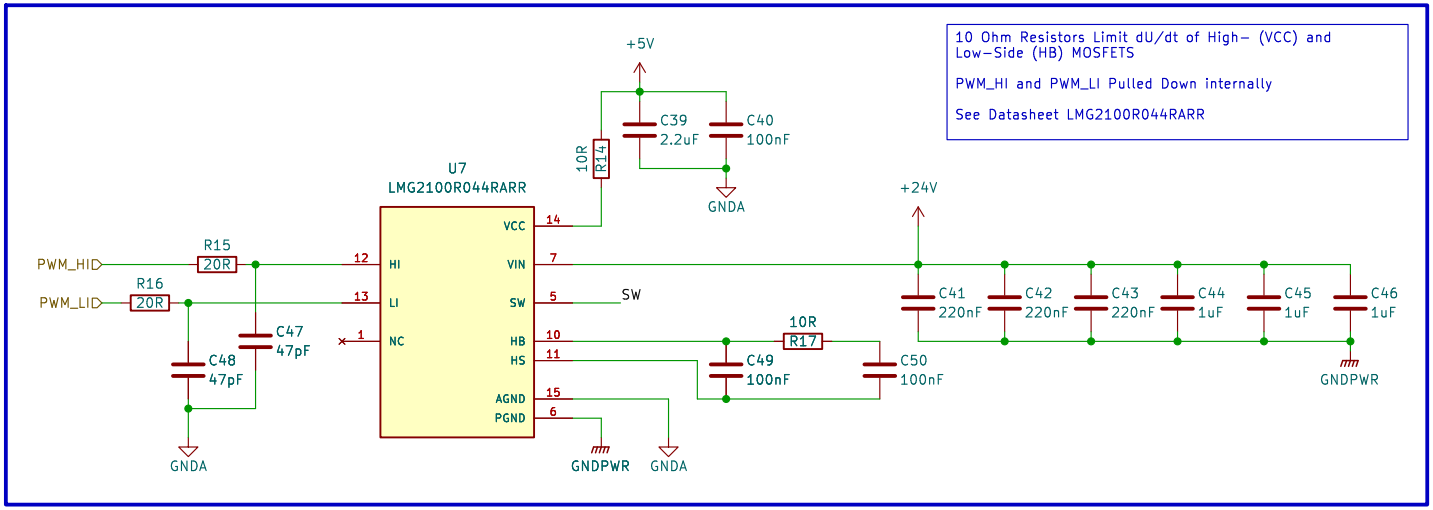
LMG2100R044 Half-Bridge Module
Uses GaN-Mosfets for fast switching



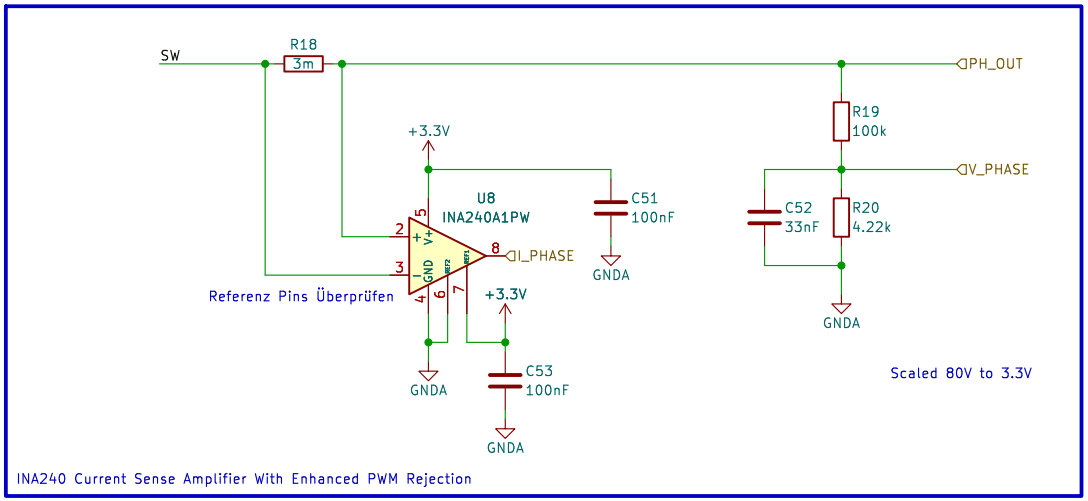
Inline Phase Shunt + OPAMP Phase Current Sensing
Voltage Divider for Phase Voltage Sensing



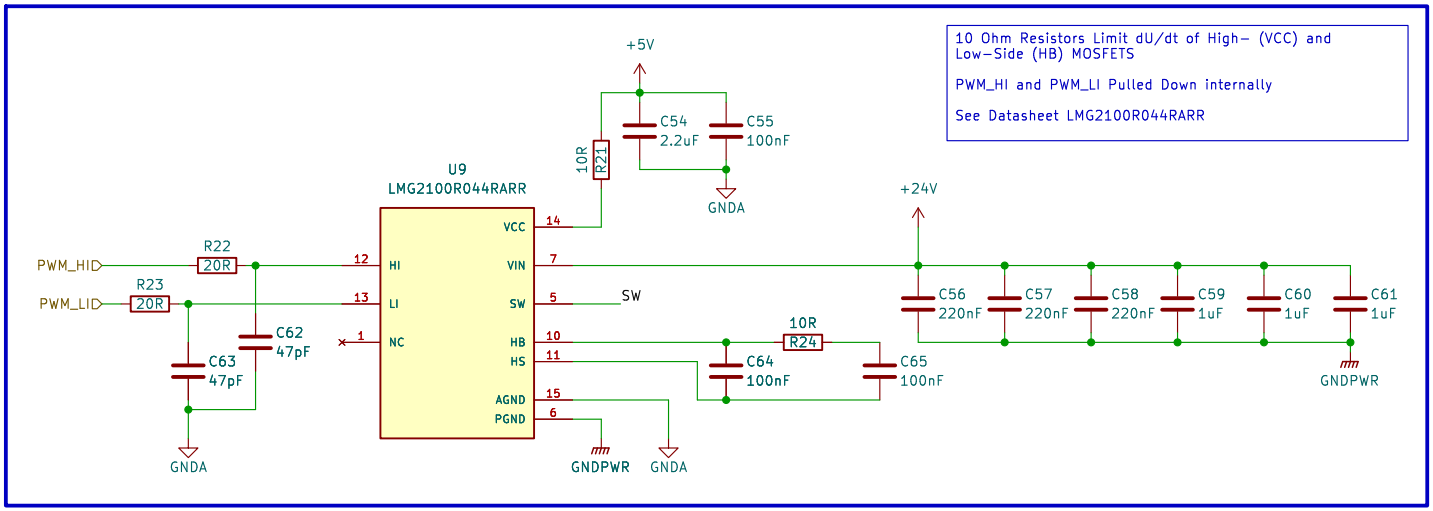
LMG2100R044 Half-Bridge Module
Uses GaN-Mosfets for fast switching



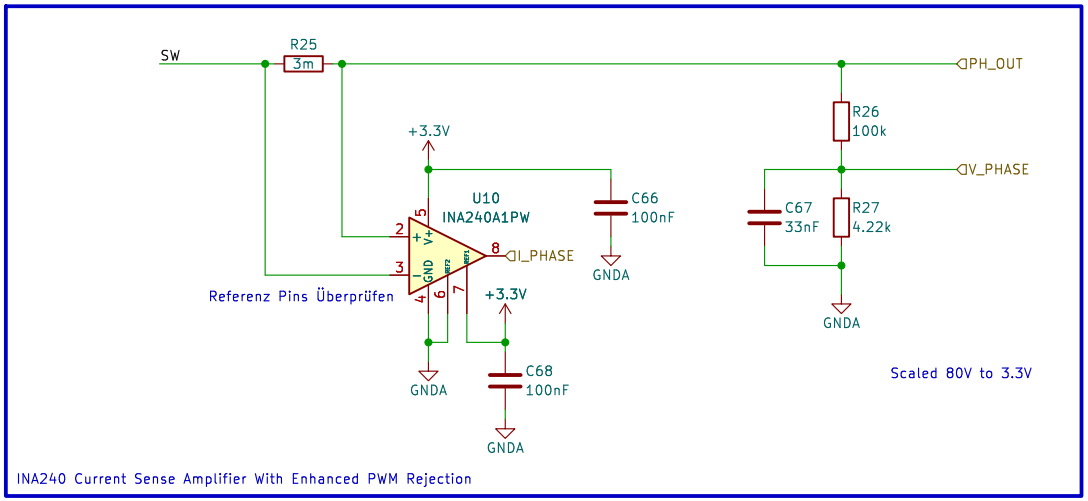
Inline Phase Shunt + OPAMP Phase Current Sensing
Voltage Divider for Phase Voltage Sensing



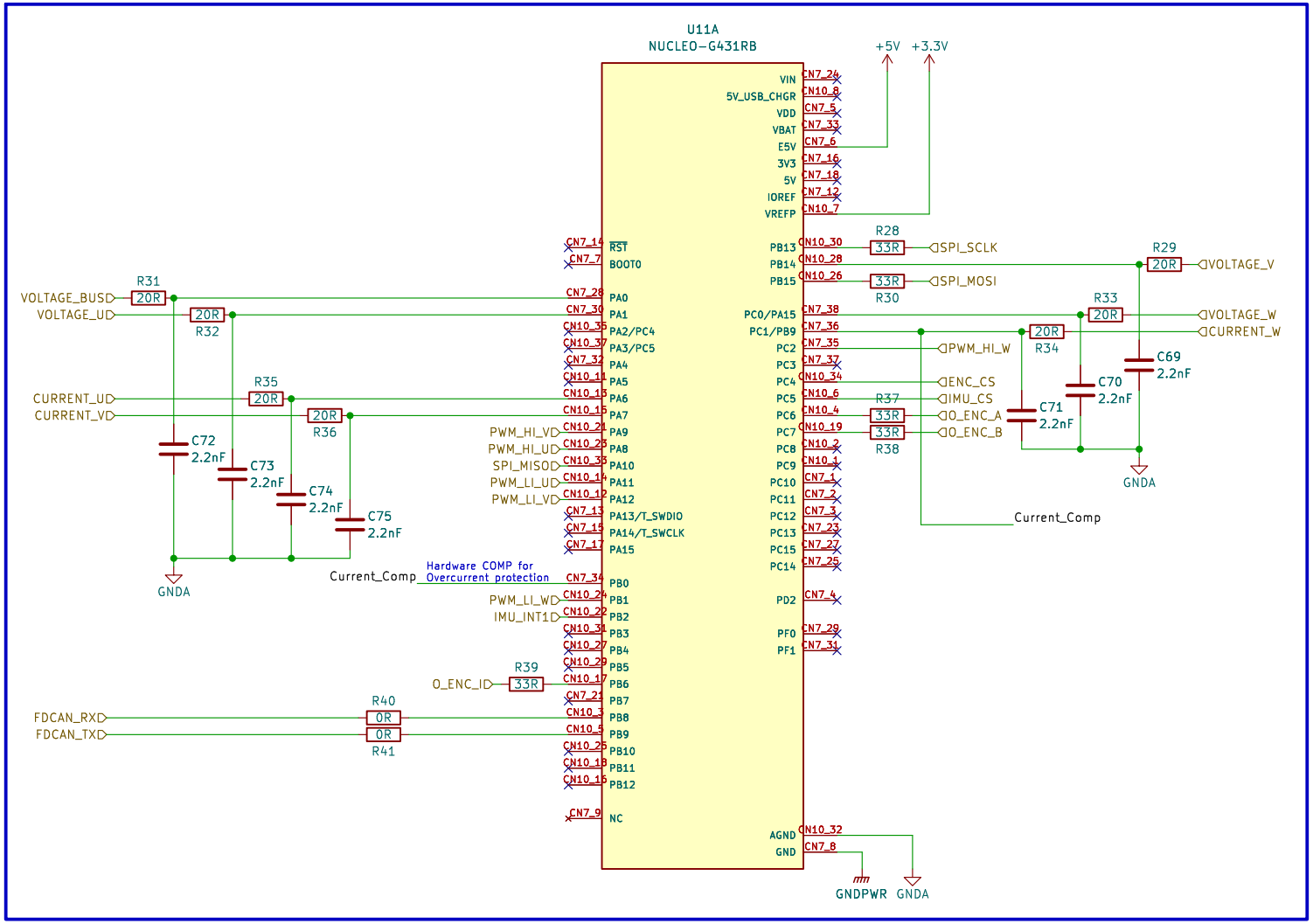
LMG2100R044 Half-Bridge Module
Uses GaN-Mosfets for fast switching



Inline Phase Shunt + OPAMP Phase Current Sensing
Voltage Divider for Phase Voltage Sensing



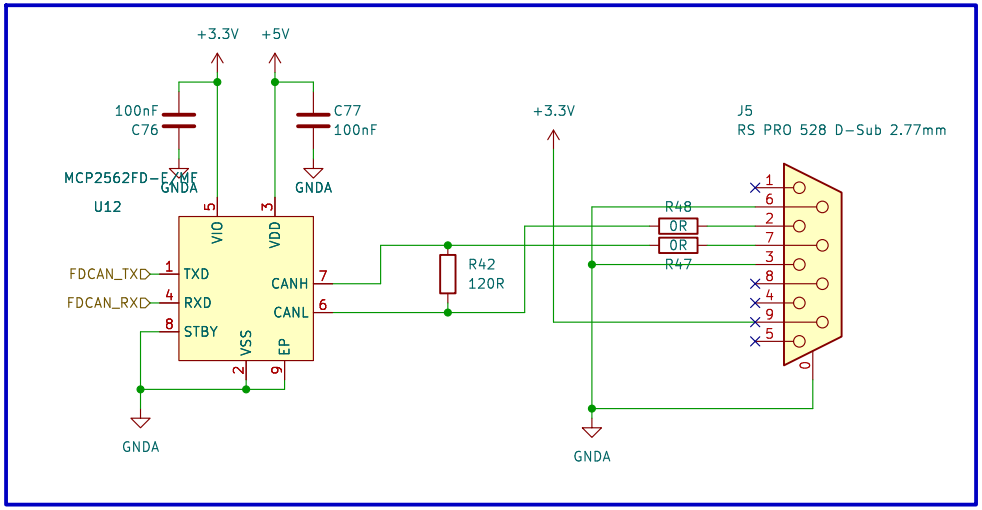
STM32 Morpho Header



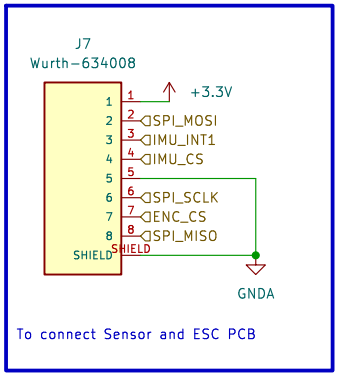
33 Ohm Resistors slow down dU/dt of MOSI and SCLK
Place close to driver

ADC Input RF filters used for ADC source impedance conditioning,
Sample capacitor charge reservoir and impulse noise damping
NOT actual analog filtering

FDCAN Transceiver + DE9 Connector



8-Pin 2.54mm Connector



6-Pin 2.54mm Connector

